

MODULE 14 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

Databases

Relational, key-value, warehouse, graph and cache. Purpose-built, and tested that way.

DOMAIN 3 of 4 EXAM WEIGHT 34% MODULE 14 of 26

Two questions in order: is it **relational**?
Then, what **shape** is the data?

The database line-up

SERVICE	THE SHAPE IT SUITS
Amazon RDS	Relational. Six engines, managed for you
Amazon Aurora	Relational. MySQL and PostgreSQL compatible, higher performance
Amazon DynamoDB	Key-value and document. Serverless, single-digit millisecond
Amazon Redshift	Analytics over a data warehouse. Columnar, petabyte scale
Amazon ElastiCache	In-memory cache in front of a slower database
Amazon Neptune	Graph. Relationships between things are the point
Amazon DocumentDB	Document. MongoDB-compatible workloads

ON THE EXAM

The exam rarely asks what a database does. It asks which one fits a described workload.

Amazon RDS or a database on EC2?

AMAZON RDS

AWS runs the database

- AWS patches the OS and the engine
- Automated backups and point-in-time recovery
- Multi-AZ failover is a checkbox
- Less control over the engine configuration

DATABASE ON AMAZON EC2

You run the database

- You patch everything and configure backups
- Full control, including unsupported engines and versions
- You build high availability yourself
- Chosen when RDS cannot meet a specific need

THE TIEBREAKER

RDS is the default. Choose EC2 only when the scenario names something RDS cannot do, such as an unsupported engine.

Amazon RDS

- ◆ **Six engines:** MySQL, PostgreSQL, MariaDB, Oracle, SQL Server and Db2.
- ◆ Existing applications usually work unchanged, because the engine is the same one they already use.
- ◆ **Automated backups** with point-in-time recovery, plus manual snapshots you keep as long as you like.
- ◆ Storage can be encrypted with KMS. Encryption is chosen at creation.
- ◆ You have no operating system access. If a scenario needs that, RDS is eliminated.

ON THE EXAM

Migrating an Oracle or SQL Server application without changing it is the classic RDS scenario.

Aurora or standard RDS?

AMAZON AURORA

AWS's own engine

- MySQL and PostgreSQL compatible only
- Higher throughput than the standard engines
- Six copies of data across three Availability Zones
- Aurora Serverless scales capacity automatically

STANDARD RDS ENGINES

The engine you already run

- Oracle, SQL Server, Db2 and MariaDB as well
- The only option for those four engines
- Familiar behaviour and tooling
- Lower cost at small scale

THE TIEBREAKER

Oracle or SQL Server means standard RDS. Aurora is only ever MySQL or PostgreSQL compatible.

Multi-AZ or read replicas?

MULTI-AZ DEPLOYMENT

For availability

- A standby copy in another Availability Zone
- Synchronous replication
- Automatic failover if the primary fails
- The standby is for resilience, not for reads

READ REPLICAS

For performance

- Additional readable copies of the database
- Asynchronous replication
- No automatic failover
- Can be created in another Region

THE TIEBREAKER

Survive a failure means Multi-AZ. Too many read queries slowing the database means read replicas.

Amazon DynamoDB

- ◆ **Key-value and document database**, fully serverless. There is nothing to provision or patch.
- ◆ **Single-digit millisecond performance at any scale**, which is its defining claim.
- ◆ On-demand capacity charges per request; provisioned capacity is cheaper for steady predictable load.
- ◆ Global tables replicate across Regions. DynamoDB Accelerator adds an in-memory cache for microsecond reads.
- ◆ No joins and no SQL. Complex relational queries are the wrong fit.

ON THE EXAM

Single-digit millisecond at any scale and serverless together are the DynamoDB signature phrases.

DynamoDB or Redshift?

AMAZON DYNAMODB

Many small, fast operations

- Look up one item by its key, constantly
- Serves an application in real time
- Milliseconds per request
- Transactional, operational workload

AMAZON REDSHIFT

Few large, complex queries

- Aggregate millions of rows for a report
- Serves analysts and dashboards
- Seconds to minutes per query
- Analytical workload, columnar storage

THE TIEBREAKER

Serving an application means DynamoDB. Answering a business question over historical data means Redshift.

Amazon Redshift

- ◆ **A columnar data warehouse** built for analytical queries over very large datasets.
- ◆ Columnar storage means a query reading three columns does not read the other ninety-seven.
- ◆ **It uses SQL**, so analysts and existing business intelligence tools work with it directly.
- ◆ Redshift Serverless removes cluster management. Redshift Spectrum queries data held in Amazon S3.
- ◆ It is not for serving an application. Do not choose it for high-volume single-row lookups.

ON THE EXAM

Data warehouse, business intelligence, reporting and analytics all point at Redshift regardless of the rest of the scenario.

Amazon ElastiCache

- ◆ **An in-memory cache** that sits in front of a database to serve repeated reads in microseconds.
- ◆ The classic use is relieving a database whose read load is dominated by the same few queries.
- ◆ **Engines are Valkey, Redis OSS and Memcached.** Valkey support was added in October 2024.
- ◆ Also used for session storage, leaderboards and rate limiting.
- ◆ A cache is not a database. Anything that must persist belongs somewhere durable.

ON THE EXAM

Repeated identical reads slowing a database means ElastiCache. Different queries every time means read replicas.

Neptune and DocumentDB

- ◆ **Amazon Neptune is a graph database.** The relationships between items are what you query.
- ◆ Recommendation engines, fraud detection rings, social networks and knowledge graphs are the named use cases.
- ◆ **Amazon DocumentDB is a document database** with MongoDB compatibility.
- ◆ The signal is an existing MongoDB application the company does not want to rewrite.
- ◆ Both are managed, so AWS handles patching, backups and availability.

ON THE EXAM

Recommendation engine or relationships between entities means Neptune. MongoDB named means DocumentDB.

What has changed here

- ◆ **Amazon QLDB reached end of support on 31 July 2025.** The ledger database is gone.
- ◆ AWS's migration path was to Amazon Aurora PostgreSQL. Older banks still offer QLDB as the ledger answer.
- ◆ **Amazon Keyspaces and Amazon MemoryDB are not on the CLF-C02 in-scope list.** Recognise, do not study.
- ◆ ElastiCache is now commonly described as Valkey rather than Redis, after the 2024 licence change.
- ◆ Everything else in this module is current and examinable.

ON THE EXAM

If a scenario describes an immutable, cryptographically verifiable ledger, the old answer was QLDB. That service no longer exists.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Migrate an Oracle application without changing it	Amazon RDS
Key-value lookups, single-digit milliseconds, any scale	Amazon DynamoDB
Business intelligence over years of historical data	Amazon Redshift
Product recommendation engine, relationships	Amazon Neptune
Existing MongoDB workload	Amazon DocumentDB
Repeated reads are overloading the database	Amazon ElastiCache
Survive the failure of an Availability Zone	RDS Multi-AZ deployment
Reporting queries are slowing the application	RDS read replicas

ON THE EXAM

Eight phrasings covering most database questions. The last two are the pair that decides the most marks.

A company runs a critical order-processing system on Amazon RDS for PostgreSQL. A recent Availability Zone disruption made the database unreachable for two hours, halting orders.

The company wants the database to keep serving requests if a single Availability Zone becomes unavailable, with failover happening automatically.

What should the company configure?

- ☐ A A read replica in another Availability Zone
- ☐ B A Multi-AZ deployment
- ☐ C Automated backups with point-in-time recovery
- ☐ D Amazon ElastiCache in front of the database

◆ PAUSE AND DECIDE

What should the company configure?

- A

 A read replica in another Availability Zone
The strongest distractor. A replica is in another AZ, but it is asynchronous and failover is manual, not automatic.
- B

 A Multi-AZ deployment
Synchronous standby with automatic failover. That is precisely what the requirement describes.
- C

 Automated backups with point-in-time recovery
Recovery after data loss, measured in hours. It does not keep the database serving requests.
- D

 Amazon ElastiCache in front of the database
Speeds up repeated reads. If the database is unreachable, writes fail regardless of the cache.

THE SIGNAL

Automatically is the decisive word. Read replicas require someone to promote them; Multi-AZ fails over on its own.

The same order-processing database is now highly available, but the finance team has begun running large reporting queries against it during the working day. Order processing has become noticeably slower whenever those reports run.

The reports can tolerate data that is a few seconds out of date.

What should the company do?

- ☐ A Add a second Multi-AZ standby
- ☐ B Create a read replica and point the reporting queries at it
- ☐ C Increase the size of the primary instance
- ☐ D Move the database to Amazon DynamoDB

◆ PAUSE AND DECIDE

What should the company do?

- A

 Add a second Multi-AZ standby
Standbys exist for failover and do not serve read traffic. It would add cost and change nothing.
- B

 Create a read replica and point the reporting queries at it
Reporting load moves off the primary, and a few seconds of lag is acceptable here.
- C

 Increase the size of the primary instance
Buys headroom without separating the workloads. The reports still compete with order processing.
- D

 Move the database to Amazon DynamoDB
Re-architecting a relational system to fix a reporting problem, and DynamoDB is poor at analytical queries anyway.

THE SIGNAL

Can tolerate data a few seconds out of date is written specifically to make asynchronous replication acceptable.

A retailer wants to suggest products by finding customers who bought similar combinations of items, then following those connections several steps outward. The queries are about how items and customers connect to one another.

Existing relational queries doing this require many joins and have become too slow.

Which database is designed for this?

- | | |
|--|--|
| ☐ A Amazon Redshift | ☐ B Amazon DynamoDB |
| ☐ C Amazon Neptune | ☐ D Amazon DocumentDB |

◆ PAUSE AND DECIDE

Which database is designed for this?

A Amazon Redshift

Excellent at aggregating rows. Traversing connections several steps outward is not what columnar analytics does.

C Amazon Neptune

A graph database. Relationships are the primary structure, so multi-step traversal is the normal operation.

B Amazon DynamoDB

Fast for looking up a known key. It has no way to traverse relationships between items.

D Amazon DocumentDB

Stores documents well, but relationships between documents are not what it queries.

THE SIGNAL

Following connections outward, and many joins being too slow, are both graph signals. Recommendation engine confirms it.

A news website stores articles in Amazon RDS. Analysis shows that ninety percent of database reads are requests for the same twenty popular articles, repeated thousands of times an hour, and the database is struggling under the load.

The articles change rarely.

What is the most appropriate way to reduce database load?

- ☐ A Add read replicas
- ☐ B Add Amazon ElastiCache in front of the database
- ☐ C Migrate to Amazon Redshift
- ☐ D Enable Multi-AZ

◆ PAUSE AND DECIDE

What is the most appropriate way to reduce database load?

A Add read replicas

Would work, but it spreads the same repeated query across more database instances rather than answering it once.

C Migrate to Amazon Redshift

A data warehouse serving a website's article reads. Wrong workload type entirely.

B Add Amazon ElastiCache in front of the database

Twenty items that rarely change, requested constantly. Cached in memory, they are served without touching the database.

D Enable Multi-AZ

Availability, not performance. The standby serves no reads.

THE SIGNAL

The same small set of items requested repeatedly is the cache signature. Varied queries would point at read replicas instead.

A gaming company needs to store player profiles and session state. It expects unpredictable growth from thousands to potentially millions of players, needs consistent single-digit millisecond reads, and does not want to manage database servers or plan capacity. Each profile is retrieved by player identifier. There are no complex joins.

Which database should the company use?

- | | |
|---|--|
| ☐ A Amazon RDS for MySQL | ☐ B Amazon Aurora |
| ☐ C Amazon DynamoDB | ☐ D Amazon Redshift |

◆ PAUSE AND DECIDE

Which database should the company use?

- A

 Amazon RDS for MySQL
Relational strengths are wasted with no joins, and you still choose an instance size and plan capacity.
- B

 Amazon Aurora
Closer, since Aurora Serverless scales, but it remains a relational engine for a pure key-value workload.
- C

 Amazon DynamoDB
Serverless, no capacity planning, single-digit millisecond reads at any scale, retrieved by key.
- D

 Amazon Redshift
Analytical warehouse. Wrong for serving an application in real time.

THE SIGNAL

Four DynamoDB signals: unpredictable scale, single-digit milliseconds, no servers to manage, retrieval by key.

What to carry out of this module

- ◆ **Relational first, then shape.** SQL and joins means the RDS family; otherwise match the data's shape.
- ◆ **Multi-AZ is availability. Read replicas are performance.** Automatic failover is the deciding word.
- ◆ Repeated identical reads means ElastiCache. Varied heavy reads means read replicas.
- ◆ Serving an application means DynamoDB. Answering a business question means Redshift.
- ◆ Aurora is MySQL or PostgreSQL compatible only. Oracle and SQL Server mean standard RDS.

ON THE EXAM

QLDB reached end of support in July 2025. If a bank offers it for a ledger workload, the service no longer exists.